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PYTHON

Artificial Intelligence Lab 03

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Lab No 03

1. **Implement** a Python program that stores 6 fruit-price pairs as tuples inside a list, e.g. `("Apple", 150)`. The program must: (a) loop through the list and print each fruit with its price, (b) use a **list comprehension** to build a new list containing only the fruit names, (c) use a list comprehension to calculate and print the total price of all fruits, and (d) demonstrate *writing over* a tuple by correcting one fruit's price (assume it was entered wrong) with a brand-new tuple, then print the updated list.

```
Input:
[("Apple", 150), ("Banana", 80), ("Mango", 200),
 ("Grapes", 250), ("Orange", 100), ("Papaya", 90)]

Sample Output:
Apple: 150
Banana: 80
Mango: 200
Grapes: 250
Orange: 100
Papaya: 90
Fruit names: ['Apple', 'Banana', 'Mango', 'Grapes', 'Orange', 'Papaya']
Total price: 870
Updated list: [('Apple', 150), ('Banana', 80), ('Mango', 180),
('Grapes', 250), ('Orange', 100), ('Papaya', 90)]
```

Code:

```python
fruits = [("Apple",150),
          ("Banana", 80),
          ("Mango", 200),
          ("Grapes",250 ),
          ("Orange", 100),
          ("Papaya", 90)
         ]

#a
for fruit in fruits:
    print(fruit)

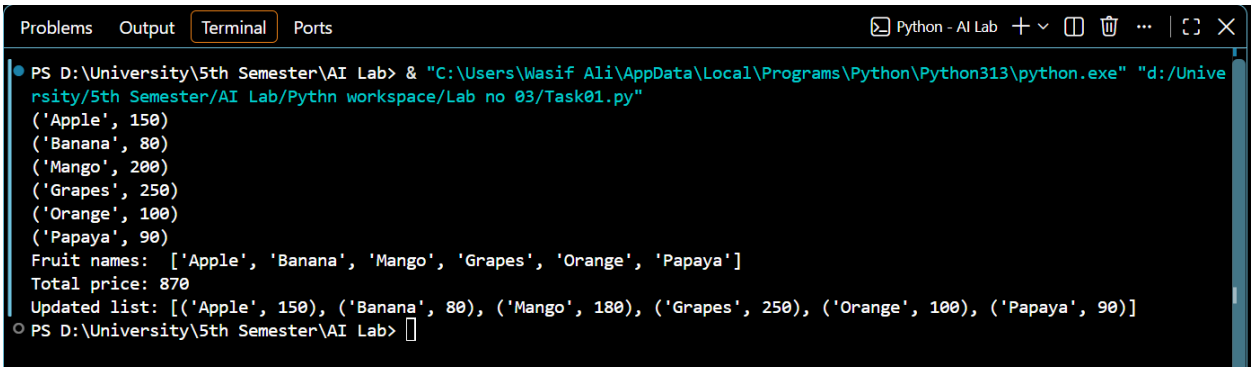
#b
names = [fruit[0] for fruit in fruits]
print("Fruit names: ", names)

#c
```

```python
prices = [fruit[1] for fruit in fruits]
print("Total price:", sum(prices))

#d
fruits[2] = ("Mango", 180)
print("Updated list:",fruits)
```

Output:



2. **Apply** dictionary operations in a Python program that stores a single contact's details in a dictionary with keys `"name"`, `"phone"`, and `"city"`. The program must: (a) print the dictionary, (b) add a new key `"email"` with a value, (c) modify the value of `"city"`, (d) remove the `"phone"` key using `pop()` and print the value that was removed, (e) use `get()` to safely check for an optional `"company"` key, printing `"Freelancer"` when it is missing, and (f) loop through the final dictionary using `items()` to display every remaining key-value pair.

```
Input:
{"name": "Hamza", "phone": "0300-1234567", "city": "Islamabad"}

Sample Output:
{'name': 'Hamza', 'phone': '0300-1234567', 'city': 'Islamabad'}
After adding email: {'name': 'Hamza', 'phone': '0300-1234567',
'city': 'Islamabad', 'email': 'hamza@example.com'}
After modifying city: Rawalpindi
Removed phone: 0300-1234567
Company: Freelancer
name -> Hamza
city -> Rawalpindi
email -> hamza@example.com
```

Code:

```python
contact = {"name": "Hamza",
           "phone": "0300-1223123",
           "city": "Islamabad"
          }

#a
print(contact)

#b
contact["email"] = "245187@aack.au.edu.pk"
print("After adding email:" , contact)

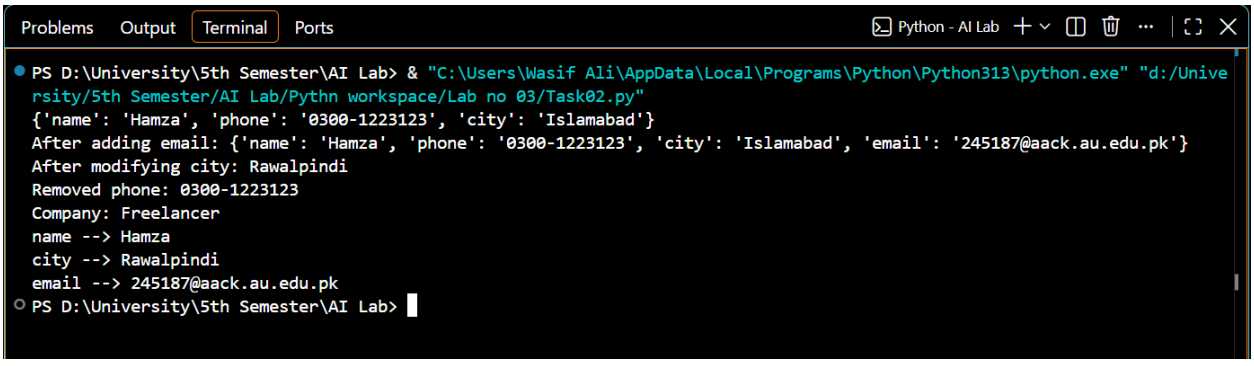
#c
contact["city"]= "Rawalpindi"
print("After modifying city:" , contact["city"])

#d
removed_phone = contact.pop("phone")
print("Removed phone:", removed_phone)

#e
print("Company:", contact.get("company","Freelancer"))

#f
for key, value in contact.items():
    print(key, "-->",value)
```

Output:



3. **Implement** a Python program to manage a small library catalog using a **list of dictionaries**, where each dictionary represents a book with `"title"`, `"author"`, and `"copies_available"` keys; some books may also have an optional `"rating"` key. The program must: (a) loop through the list and display each book's title and author, (b) use a **list comprehension** to build a list of titles for books that currently have at least 1 copy available, (c) use `get()` to safely retrieve each book's `"rating"`, displaying `"Not Rated"` when it is missing, and (d) update the `"copies_available"` count for one book after it is borrowed (decrement by 1), then print the updated catalog.

```
Input:
[
 {"title": "Introduction to AI", "author": "Stuart Russell",
  "copies_available": 3, "rating": 4.8},
 {"title": "Python Crash Course", "author": "Eric Matthes",
  "copies_available": 0},
 {"title": "Deep Learning", "author": "Ian Goodfellow",
  "copies_available": 5, "rating": 4.6}
]

Sample Output:
Introduction to AI by Stuart Russell
Python Crash Course by Eric Matthes
Deep Learning by Ian Goodfellow
Available titles: ['Introduction to AI', 'Deep Learning']
Introduction to AI rating: 4.8
Python Crash Course rating: Not Rated
Deep Learning rating: 4.6
After borrowing "Introduction to AI", copies available: 2
Updated catalog: [{'title': 'Introduction to AI', 'author': 'Stuart Russell'
    ,
'copies_available': 2, 'rating': 4.8}, {'title': 'Python Crash Course',
'author': 'Eric Matthes', 'copies_available': 0}, {'title': 'Deep Learning',
'author': 'Ian Goodfellow', 'copies_available': 5, 'rating': 4.6}]
```

Code:

```python
catalog = [
{"title": "Introduction to AI", "author": "Stuart Russell",
"copies_available": 3, "rating": 4.8},
{"title": "Python Crash Course", "author": "Eric Matthes",
"copies_available": 0},
{"title": "Deep Learning", "author": "Ian Goodfellow",
"copies_available": 5, "rating": 4.6}
]

#a
for book in catalog:
    print(book["title"],"by", book["author"])
```

```python
#b
avaible_titles = [book["title"] for book in catalog if book["copies_available"]
>= 1]
print ("Available titles:", avaible_titles)

#c
for book in catalog:
    print(book["title"], "rating:", book.get("rating", "Not Rated"))

#d
catalog[0]["copies_available"]-=1
print('Copies available:', catalog[0]["copies_available"])
print("Updated catalog:", catalog)
```

Output:

